Paz-Pascual *et al. BMC Pregnancy and Childbirth* (2025) 25:682
https://doi.org/10.1186/s12884-025-07803-w

BMC Pregnancy and Childbirth

RESEARCH Open Access

Analysis of variables affecting postpartum quality of life: their relative influence and interrelationships

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Abstract

Background The postpartum period is a time of special vulnerability and unacceptably high morbidity for the mother, which warrants in-depth study. The objective of this study is to analyze the variables affecting quality of life (QoL) during the first six months postpartum, their relative influence, and the relationships between them.

Methods Cross-sectional study with the participation of 289 postpartum women, who answered the digital questionnaires: SF-12 for QoL; EMAeHealth for self-management of health in the postpartum, which assesses: incontinence, sexuality, breastfeeding, adaptation to the maternal role and mental health; and "Prescribe Healthy Life" Screening Questionnaire, which measures diet, exercise and consumption of toxic. To analyze the relationship between the variables and their relative influence on QoL, a structural equation model was built.

Results Depression (β=-0.640; 90%CI: (-0.670)-(-0.505); $p=0.006$), partner support (β = 0.331; 90%CI:0.217–0.424; $p=0.003$), parental self-efficacy (β = 0.219; 90%CI:0.134–0.309; $p=0.001$) and physical QoL (β=-0.223; 90%CI: (-0.294)-(-0.142); $p=0.002$) are related to the mental dimension of QoL. Partner support also influences the breastfeeding self-efficacy (β = 0.28; 90%CI:0.197–0.373; $p=0.002$), sexual satisfaction (β = 0.48; 90%CI:0.38–0.56; $p=0.03$), body dissatisfaction (β=-0.155; 90%CI: (-0.27)-(-0.04); $p=0.031$), and parental self-efficacy (β = 0.170; 90%CI:0.027–0.344; $p=0.045$), which indirectly affect QoL. Depressive symptoms (β=-0.126; 90%CI: (-0.218)-(-0.026); $p=0.044$) and smoking (β=-0.224; 90%CI: (-0.314)-(-0.132); $p=0.002$) affect the physical component of QoL.

Conclusions This study explains the influence of the variables involved in QoL after childbirth and their interrelationship, providing a graphical representation that facilitates a comprehensive understanding. This knowledge forces us, as midwives, to review perinatal and postpartum care with the aim of paying special attention to partner support and depressive symptoms.

Keywords Quality of life, Postpartum, Physical quality of life, Mental quality of life

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Introduction

After childbirth, a period of particular vulnerability for the mother begins, marked by physical, psychological and social changes. These changes mean a greater predisposition to suffering certain health problems and the need to face new emotional and psychological challenges [1].

If we start from the assumption that the woman is at the center of a dynamic network of interactions between factors that determine her physical and emotional health, then to focus only on the biomedical causes of this maternal vulnerability seems insufficient. Among health factors, we should take into account those related to the economic, political and cultural situation of the society in which each woman lives; social factors, such as gender, income, education, ethnic origin and race; and individual factors specific to women in general [2, 3]. Individual-level factors are related to the specific characteristics of each woman, such as her age, genetics, pre-existing health conditions, exposure to external agents, and individual factors shaped by family and community characteristics, such as diet, physical activity, substance abuse, and certain sexual behaviors [2].

To this entire catalog of factors that influence health, we must add the experience of pregnancy and childbirth, with processes that can cause physical and emotional sequelae that cause adverse outcomes in the postpartum period [4, 5]. Although most pregnancies and births occur without incident, all pregnancies present risks [6] and even a vaginal birth with a normal progression can have postpartum complications [7]. While the vast majority of interventions performed during childbirth that are offered to minimize harm to the mother and baby are appropriate and justified, they can cause iatrogenic complications and trigger negative physical, social, or psychological results that interfere with the woman's recovery. Examples include episiotomy, instrumental vaginal delivery and cesarean Sects. [4, 7]. Another factor to consider is the context of the delivery; giving birth in an unfamiliar, hostile or unsupportive clinical environment affects a woman's emotional adaptation to motherhood and is associated with adverse long-term clinical and emotional outcomes, such as feelings of competence, self-confidence and initiation of breastfeeding after delivery [7].

Most of the approximately 140 million women who give birth each year worldwide experience postpartum morbidity [7, 8]. The most common conditions include dyspareunia, (35%), lower back pain (32%), urinary incontinence (8–31%), anxiety (9–24%), anal incontinence (19%), depression (11–17%), tokophobia (intense fear of childbirth) (6–15%), perineal pain (11%) and secondary infertility (11%)[7].

This data does not seem acceptable if, after the care received by women's and maternity care services, the expected result is good quality of life for the woman in the postpartum period; if we see quality of life as a multidimensional concept that is affected by physical, psychological, relational and social well-being [9, 10] and is considered valuable for women during motherhood [11].

Many of the variables described, involved in health and quality of life, are difficult for the health system to modify, since they affect social or economic areas; however, others could be addressed by primary care. We know that sleep deprivation, difficulty in breastfeeding and poor support in care can increase the risk of postpartum depression [7]and that sexual satisfaction, fatigue, mental health, success in breastfeeding, confidence in the mother's role, mother-child attachment and social support significantly influence the quality of life of women after childbirth [9, 12–14]. However, there is a lack of information on how all these factors relate to each other, which ones have the greatest weight and which have a significant influence, even if indirectly, on quality of life.

The objective of this study is, therefore, to evaluate the influence of the multiple variables that in principle would affect the health of the postpartum woman in the first six months [13]analyze the interaction between them, and identify the variables that have a greater impact, directly or indirectly, on the quality of life of a woman in the postpartum period. This will allow for the selection of those that are potentially modifiable early on, and the design of interventions that improve the health and well-being of women in this stage of transition to motherhood.

Methods

The data from this study is part of a broader research that analyzes the perceptions and needs of women during pregnancy, childbirth and postpartum, and the resources available to them to adapt to the new situation. This project developed and evaluated the validity and reliability of two self-administered analysis instruments for the evaluation of perinatal psychosocial needs and health and well-being in the postpartum period. The study protocol has been published previously [15]. The Clinical Research Ethics Committee of the Basque Country (Spain) approved this study (Ref: No. PI2019110).

This is a cross-sectional study carried out in the Basque Health Service, a public service that provides care to a population of just over two million inhabitants. It operates 7 hospitals where pregnant women receive assistance during childbirth from qualified professionals, midwives, and obstetrician-gynecologists. Each hospital is coordinated with a set of primary care centers for pregnancy and postpartum follow-up. The low-risk pregnancy and postpartum care program includes 8–10 prenatal visits and at least two visits within the first 6 weeks postpartum.

Participants

The women in the study were recruited by the midwife at the postpartum check-up consultation consecutively rather than randomly. All women who met the inclusion criteria were invited to participate between August 2020 and May 2022 between August 2020 and May 2022. They were offered the possibility of receiving the link to the EMAe-Health questionnaire, for self-management of health and well-being in the postpartum period in digital format, and were encouraged to share the link with other women in the same situation. All postpartum women over 18 years of age who had sufficient command of Spanish to understand and answer the questions presented were included. When the woman went into the questionnaire, she received information about the characteristics of the study, the type of use that would be made of the data (for research purposes only) and about the possibility of withdrawing from the study at any time without compromising her standard of care. If the woman agreed to the informed consent, she would then have access to the questionnaire.

Since the information to be gathered requires a certain amount of privacy, an online questionnaire was considered the best option for data collection, since it can be completed at any time and place. The platform *encuestafacil.com* was selected to host the questionnaire, because it offers security in data protection following European Regulations. The data was transferred to an Excel format and subsequently anonymized by the data manager before proceeding with database cleaning and analysis.

The quality of the participant recruitment process and the management and analysis of the data was continuously monitored by the research team and experts from the Biocruces Bizkaia Research Institute.

Table 1 Dimensions assessed in the EMAe-Health questionnaire for self-management of health and well-being in the postpartum period

EMAe-Health Dimensions		Item numbers
Incontinence		1
Sexuality:	Pain with sexual intercourse	4
	Satisfaction with sexual intercourse	6
	Contraception	1
Breastfeeding	Knowledge about benefits	6
	Expectations and intentions about the practical aspects	13
	Perceived self-efficacy	6
Adaptation to the maternal role	Parental self-efficacy	11
	Perceived social support	6
Mental health	Body image	8
	Sleep problems	6
	Depression	6

Variables and measurement instruments

In order to determine the characteristics of the sample, information was collected regarding sociodemographic variables: age, nationality (Spanish/immigrant); level of education (primary/secondary/university); parity (primiparous/multiparous); employment status (paid work yes/no) and "type of delivery" with three options (eutocic, instrumental or cesarean).

The EMAe-Health questionnaire was used; it is a questionnaire for self-management of health and well-being in the postpartum period. It has been validated for the Spanish population [16] and has shown good metric properties in all its dimensions, with a good fit for the initial theoretical models and high internal consistency. It includes the following dimensions: incontinence, sexuality, breastfeeding, adaptation to the maternal role and mental health (Table 1). These dimensions are measured with items that are answered on a 5-option Likert-type scale.

The variables related to self-care were collected using the "Prescribe Healthy Life" Screening Questionnaire (PVS-SQ) [17]. This questionnaire is validated for the Spanish population and include 12 multiple-choice items, of which 6 measure physical activity, 2 measure fruit and vegetable consumption, 1 is about tobacco use and 3 refer to other aspects.

The outcome variable was health-related quality of life [18] measured with the Spanish version of the SF-12 v2 health questionnaire [19]. The SF-12 scale is a self-administered tool that was developed from the SF-36 scale and has 12 questions divided into 8 dimensions: physical function (2 questions), physical role (2 questions), bodily pain (1 question), general health (1 question), vitality (1 question), emotional role (2 questions), social function (1 question) and mental health (2 questions), which as a whole give rise to the physical and mental components of health-related quality of life.

Data analysis

Missing data represented a small percentage of the dataset and showed no identifiable pattern, suggesting that data were missing at random (MAR). To address this, the Full Information Maximum Likelihood (FIML) procedure implemented in AMOS was applied prior to statistical analysis. This method allows for parameter estimation using all available data without the need for direct imputation, and is widely recommended for path analysis under MAR conditions, as it reduces potential bias and preserves statistical power.

Following the treatment of missing data, descriptive statistics were calculated, including frequency, mean, and standard deviation. To establish the relationship between the different variables and Quality of Life (QoL), variance analyses were carried out in the case of qualitative

variables (ANOVA) and the raw mean difference (MD) of the scores was calculated by means of linear regressions in the case of quantitative variables. Finally, Structural Equation Modelling was applied to pathway mapping and measurement accuracy of the impact of the potentially modifiable variables on QoL. A P-value of < 0.05 was considered statistically significant.

Several models were tested to determine the relationships between the variables that were found to have significant associations with postpartum health-related quality of life in previous analyses. To compare the different alternative models, firstly the full model was estimated with all the variables of interest. The fit of this model was considered the baseline model for comparison for other simpler models that might be proposed. When non-significant effects were eliminated from the model, the fit of the simplified model was compared with that of the full model. If the elimination of non-significant coefficients was relevant, the fit of the simplified model had to be at least as good as that of the baseline model.

Once the final model was specified using path diagrams, the regression coefficients were estimated from the empirical variance-covariance matrix among the observable variables. The coefficients obtained were those that best reproduced the covariances between the variables and their corresponding variances. The fit of the model was assessed from two points of view. First, the estimated coefficients were checked for significance, i.e., it had to be possible to conclude that their value was different from zero in the population. Second, the degree of coincidence between the original variance-covariance matrix and the matrix reproduced from the estimated parameters was assessed using various goodness-of-fit statistics. Given that the deviations in the distribution of scores from the normal curve are small (levels in asymmetry and kurtosis statistics below 2 and 5 respectively), the estimation method used in the analyses is the maximum likelihood (ML) method, which is robust under this condition [20]. The goodness of fit of the model was assessed using the χ2 statistic, the ratio between χ2 and the number of degrees of freedom (χ2/df), whose value had to be less than 5 to indicate an adequate fit, the GFI (Goodness of Fit Index), AGFI (Adjusted Goodness of Fit Index), NFI (Normed Fit Index), TLI (Tucker-Lewis Index), CFI (Comparative Fit Index) and IFI (Incremental Fit Index) indices, whose values had to be greater than 0.90, and the SRMR (Standardized Root Mean Square Residual) and RMSEA (Root Mean Square Error of Approximation) indices, whose values must be less than 0.08 [21, 22]. Indirect effects between variables were studied using the bootstrap method on 1000 simulated samples [23].

Results

Of the 443 women invited to participate in the consultation with the primary care midwife, 348 women answered at least one question and 216 (48.75%) completed the entire questionnaire. In addition, 64 women were recruited through informal contact between participants. Finally, 280 postpartum women with a mean age of 34.93 years (SD = 4.74) participated, and their characteristics and relationship with the scores on quality of life (SF-12) in the postpartum period can be seen in Table 2.

In the final estimated model (Fig. 1), all the fit indicators are considered good [χ [2] ($df = 69$, $N = 280$) = 132.5, $p < 0.001$; [$\chi^2/df = 1.92$; GFI = 0.937; AGFI = 0.905; NFI = 0.903; IFI = 0.951; TLI = 0.934; CFI = 0.951; RMSEA (90%CI) = 0.057 (0.043–0.256); SRMR = 0.072]. In this model, the set of predictors explains 50.9% (90%CI = 42.4–57.5; $p = 0.006$) of the variance of the scores in the mental component of the quality of life assessed by the SF-12.

As can be seen in Table 3, all the direct links between variables are statistically significant. The results of the evaluation of the direct, indirect and total effects on the mental component of quality of life are laid out in Table 4.

The most intense total effect is the negative association between depression and the mental component of quality of life (β=-0.640; 90%CI:-0.670 to -0.505), mostly in a direct way (β=-0.612; 90%CI:-0.677 to 0.523; $p = 0.005$). The path mediated by the physical component of quality of life accounts for a small part of the explained variance, and also shows an inverse relationship: the higher the depression score, the lower the quality of life (β=-0.028 90%CI: -0.055 to -0.005; $p = 0.029$).

The second most influential factor associated with quality of life is partner support (β = 0.331; 90% CI: 0.217 to 0.424; p = 0.003), which exerts a positive y direct influence (β = 0.136; 90% CI: 0.028 to 0.221; p = 0.047) and mediated influence (β = 0.195; 90% CI: 0.135 to 0.250; p = 0.002), through several pathways: (a) parental self-efficacy and more complex indirect paths involving sexual satisfaction, which is negatively associated with depression and, in turn, positively influences both the physical and mental components of quality of life; (b) body dissatisfaction, which is positively associated with pain during sexual intercourse and negatively with sexual satisfaction, both sexual satisfaction and sexual pain impact depression and, therefore, quality of life; and (c) self-efficacy for breastfeeding, which influences breastfeeding practice, which affecting perceived parental self-efficacy. This, in turn, is negatively associated with depression and positively associated with the mental component of quality of life."

In the case of body dissatisfaction, the third dimension with the greatest explanatory power, the entirety of the harmful effect it exerts on quality of life can be attributed

Table 2 Characteristics of the participants and their relationship with the physical and mental dimensions of quality of life

		N (%)	Physical QoL		Mental QoL	
			Mean (SD)	Mean diference (95%IC)*	Mean (SD)	Mean diference (95%IC)*
Parity						
	Multiparous	97 (35)	47,1 (8,99)	Ref	45,26(11,44)	Ref
	Primiparous	180 (65)	45.4 (10,32)	1.66(-0.79, 4.11)	46,4(11,53)	-1.11(-3.96, 1.74)
Nationality						
	Spanish	254(90.7)	46,03 (9,9)	Ref	45,85 (11,56)	Ref
	Immigrant	26 (9,39	46,7 (9,6)	-0.7(-4.78, 3.38)	47,01(11,42)	-1.16(-5.93,3.6)
Education						
	Primary	14 (5,05)	44,3 (9,8)	Ref	49,17(12,9)	Ref
	Secondary	89 (32.13)	43,9 (11,03)	0.34(-5.86, 6.55)	47,8(11,7)	1.32 (-5.44, 8.09)
	University	174 (62,81)	47,2 (9,13)	-2.89(-7.92, 2.13)	44,8(11,16)	4.39(-1.79, 10.58)
Paid work						
	Yes	226 (81,9)	46,03 (9,72)	Ref	45,14(11,71)	Ref
	No	50(18,1)	45,9 (10,86)	1.38(-2.92, 3.19)	49,61 (9,82)	**-4.47(-7.98, -0.97)***
Type of delivery						
	Eutocic	180(63,8)	47,31(9,66)	Ref	45,6(11,26)	Ref
	Instrumental	57 (20,2)	46,2 (9,16)	1.12(-1.75,4,01)	46,3 (11,97)	-0.67(-4.12,2.78)
	Cesarean	45 (16)	40,4(10,05)	**6.89(3.63, 10.15)***	47,01 (12,04)	-1.37(-5.2, 2.44)
Smoker						
	No	258	46,22(9.88)	Ref	45.87(11.45)	Ref
	Yes	19	43.27(10.03)	2.95(-1.67, 7.58)	47.46 (12.31)	-1.59(-6.98, 3.79)
Incontinence						
	Yes	62 (35,03)	44,32 (10,75)	Ref	43,64 (11,99)	Ref
	No	215 (64,97)	46,5 (9,61)	-2.18(-4.98, 0.62)	46,65 (11,28)	**-3(-6.26, 0.24)***
		Mean (sd)	**B coefficient**	**(95%IC)***	**B coefficient**	**(95%IC)***
Age		35,04 (4,23)	0,026	(-0,37, 0,24)	-0,039	(-0,458, 0,241)
Social support		26,93 (4,18)	-0,024	(-0,35, 0,23)	0,36	**(0,68, 1,31)***
Breastfeeding, knowledge		25,23 (4,04)	0,06	(-0,252, 0,538)	0,038	(-0,370, 0,595)
Breastfeeding, practice		52,69 (6,15)	0,087	(-0,077, 0,353)	0,329	**(0,367, 0,835)***
Breastfeeding, self efficacy		24,78 (4,03)	0,16	**(0,073, 0,721)***	0,317	**(0,526, 1,246)***
Sexuality, pain		12,64 (5,65)	-0,164	(-0,681, 0,110)	-0,243	**(-1,061, -0,044)***
Sexuality, satisfaction		23,52 (5,56)	-0,028	(-0,438, 0,342)	0,371	**(0,349, 1,297)***
Body image		17,16 (6,62)	-0,199	**(-0,469, -0,123)***	-0,328	**(-0,761, -0,373) ***
Sleeping difficulties		16,04 (4,11)	0,009	(-0,403, 0,447)	-0,381	**(-1,596, -0,641)***
Depression		13,81 (5,17)	-0,178	**(-0,576, -0,120) ***	-0,678	**(-1,736, -1,340)***
Parental self efficacy		48,47 (5,43)	0,128	**(0,016, 0,458) ***	0,318	**(0,432, 0,918) ***

Abbreviations: QoL = Quality of life. $*p \leq 0{,}005$

to indirect pathways (β = 0-0.251; 90%CI:-0.309 to -0.191; $p = 0.003$). A significant portion of this effect occurs through its direct association with depression scores, while another part operates through more complex indirect pathways that involve variables related to sexual activity.

The physical component of quality of life is significantly, although inversely, related to the mental component (β=-0.223; 90%CI:-O.294 to -0.142; $p = 002$).

Parental self-efficacy exerts influence directly (β = 0.121; 90%CI:0.024 to 0.225; $p = 0.047$) and indirectly through depression (β = 0.098; 90%CI:0.034 to 0.169; $p = 0.008$). Satisfaction with sexual activity is an indirect promoter of better quality of life (β = 0.193; 90%CI:0.137 to 0.252); $p = 0.003$) while pain during sexual intercourse (β=-0.181; 90%CI:-0.242 to -0.127; $p = 0.002$) and difficulty sleeping (β=-0.126; 90%CI:-0.199 to -0.068; $p = 0.001$) are risk factors.

The aspects with the least influence are those related to breastfeeding, urinary incontinence and smoking, despite all indirect pathways of influence being statistically significant.

The use of FIML for handling missing data did not result in substantial changes in parameter estimates or model fit, suggesting the robustness of the findings.

Discussion

The main findings of the study highlight the significant impact of depression and partner support on both the physical and mental components of quality of life during

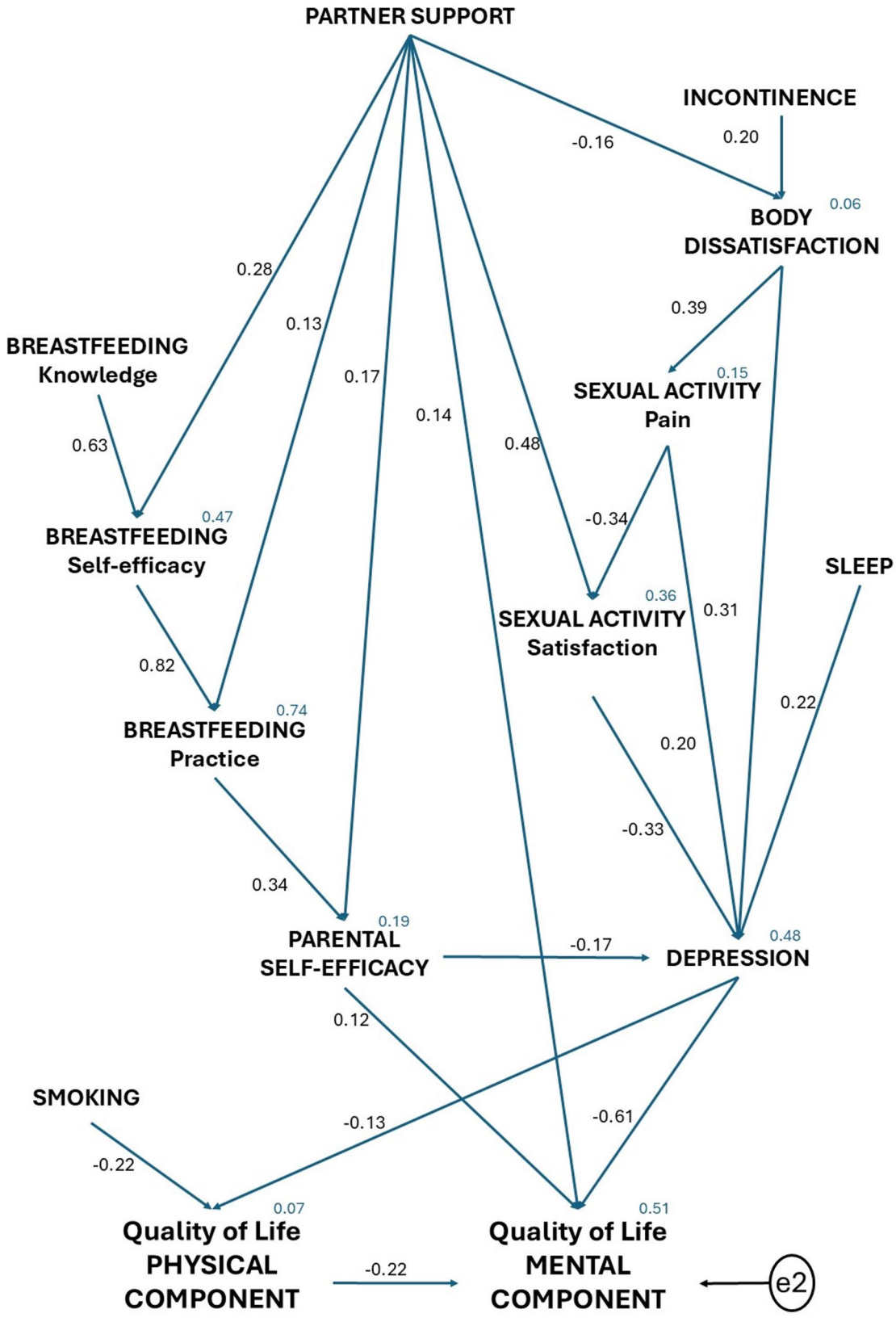

Fig. 1 Final model of relationships between variables (standardized regression coefficients) in the group of postpartum women

Table 3 Direct effects between the variables analyzed in the final explanatory model

Variables			β	LI	UI	*p*
Breastfeeding, **Self-efficacy**	<---	Partner Support	0.285	0.197	0.373	0.002
	<---	Breastfeeding, Knowledge	0.625	0.566	0.674	0.003
Breastfeeding, **Practice**	<---	Breastfeeding, Self-efficacy	0.815	0.754	0.876	0.001
	<---	Partner Support	0.127	0.061	0.207	0.003
Body Dissatisfaction	<---	Urinary Incontinence	0.196	0.094	0.299	0.004
	<---	Partner Support	-0.155	-0.266	-0.036	0.031
Sexual Activity, **Pain**	<---	Body Dissatisfaction	0.391	0.294	0.472	0.003
Sexual Activity, **Satisfaction**	<---	Partner Support	0.476	0.382	0.565	0.003
	<---	Sexual Activity, Pain	-0.338	-0.451	-0.198	0.002
Parental Self-efficacy	<---	Breastfeeding, Practice	0.341	0.213	0.478	0.002
	<---	Partner Support	0.171	0.027	0.344	0.045
Depression	<---	Sleeping Problems	0.216	0.118	0.332	0.001
	<---	Parental Self-efficacy	-0.167	-0.278	-0.068	0.007
	<---	Sexual Activity, Pain	0.198	0.105	0.298	0.002
	<---	Sexual Activity, Satisfaction	-0.331	-0.414	-0.242	0.002
	<---	Body Dissatisfaction	0.308	0.222	0.388	0.002
Physical Component SF-12	<---	Smoking	-0.224	-0.314	-0.132	0.002
	<---	Depression	-0.126	-0.218	-0.026	0.044
Mental Component SF-12	<---	Parental Self efficacy	0.121	0.024	0.225	0.047
	<---	Physical Component SF-12	0.223	0.294	0.142	0.002
	<---	Depression	-0.612	-0.677	-0.523	0.005
	<---	Partner Support	0.136	0.028	0.221	0.047

Abbreviations: QoL: Quality of Life. *$p \leq 0{,}005$B = standardized regressor coefficient, LI = Lower limit of 90% confidence interval, UI = Upper limit of 90% confidence interval, p = level of significance associated with the statistic

Table 4 Indirect, direct and total standardized effects between variables in the final model

Variables	Indirect effects β (IC90%); *p*	Direct effects β (IC90%); *p*	Total effect β (IC90%); *p*
Depression→ Mental component SF-12	−0.028 (.-0.055 to −0.005); 0.029	−0.612 (-0.677 to −0.523); 0.005	−0.640 (-0.670 to −0.505); 0.006
Partner support→ Mental component SF-12	0.195 (0.135 to 0.250); 0.002	0.136 (0.028 to 0.221); 0.047	0.331 (0.217 to 0.424); 0.003
Body dissatisfaction → Mental component SF-12	−0.251 (-0.309 to −0.191); 0.003	0.000 (0.000 to 0.000);.ne	−0.251 (-0.309 to −0.191); 0.003
Physical component SF-12→ Mental component SF-12	0.000 (0.000 to 0.000);.ne	0.223 (0.142 to 0.294); 0.002	0.223 (0.142 to 0.294); 0.002
Parental Self-efficacy → Mental component SF-12	0.098 (0.034 to 0.169); 0.008	0.121 (0.024 to 0.225); 0.047	0.219 (0.134 to 0.309); 0.001
Sexual Activity, Satisfaction → Mental component SF-12	0.193 (0.137 to 0.252); 0.003	0.000 (0.000 to 0.000);.ne	0.193 (0.137 to 0.252); 0.003
Sexual Actividad, Pain→ Mental component SF-12	−0.181 (-0.242 to −0.127); 0.002	0.000 (0.000 to 0.000);.ne	−0.181 (-0.242 to −0.127); 0.002
Sleeping Problems → Mental component SF-12	−0.126 (-0.199 to −0.068); 0.001	0.000 (0.000 to 0.000);.ne	−0.126 (-0.199 to −0.068); 001
Breastfeeding, Practice → Mental component SF-12	0.075 (0.034 to 0.132); 0.001	0.000 (0.000 to 0.000);.ne	0.075 (0.034 to 0.132); 0.001
Breastfeeding, self-efficacy → Mental component SF-12	0.061 (0.029 to 0.110); 0.001	0.000 (0.000 to 0.000);.ne	0.061 (0.029 to 0.110); 0.001
Smoking → Mental component SF-12	0.05 (0.026 to 0.084); 0.001	0.000 (0.000 to 0.000);.ne	0.050 (0.026 to 0.084); 0.001
Urinary Incontinence → Mental component SF-12	−0.049 (-0.081 to −0.023); 0.004	0.000 (0.000 to 0.000);.ne	−0.049 (-0.081 to −0.023); 0.004
Breastfeeding, knowledge→ Mental component SF-12	0.038 (0.017 to 0.070); 0.001	0.000 (0.000 to 0.000);.ne	0.038 (0.017 to 0.070); 0.001

Abbreviations: ne = not estimable

the postpartum period, exerting their influence both directly and indirectly through other variables.

The analysis of the influence of the variables that affect the quality of life of women and the relationship between them shows that the variable with the greatest weight is the presence of depressive symptoms or thoughts. This finding coincides with previous studies [12, 24, 25]which connected high scores in depression with lower quality of life. The resulting depression-related variables also agree with other studies [7, 12–14]with a significant relationship established between depression and sleep [26]body

dissatisfaction [27]dyspareunia, sexual dissatisfaction, and parental self-efficacy [7, 12–14].

Our results highlight the importance of partner support, both for its direct effect on quality of life and for the rest of the variables that ultimately affect it. The relationship between the quality of partner support and postpartum depression [28] is well known, as is partner support and self-efficacy in breastfeeding [29]including its relationship with the development of the emotional bond between mother and child [30]. However, the relationship between partner support, self-image and sexual satisfaction, which was significant in our study, is less known.

The physical changes that pregnancy and childbirth entail cause a change in a woman's self-image that can be negative and affect her emotional well-being [31]. This negative self-image affects the quality of sexual life and romantic relationship after childbirth [32]so it seems logical that the support of the partner plays a fundamental role in improving negative body image and sexual dissatisfaction. Who better than the partner to provide understanding and complicity to be able to address all these changes with trust and affection, without overlooking of course the weight of family support, which can contribute to optimal support [33]. Adequate support, satisfaction with body image and satisfaction with sexual activity can lead to a reduction in the incidence of postpartum depression, given their statistically significant relationship.

Regarding the relationship, also statistically significant, between partner support and parental self-efficacy, it seems clear that the birth of a child involves readjustments in the family system that affect both members of the couple at the same time and which both have to face, attending to the needs of the other and to the shared needs and concerns [34]. It is thus understood that a woman who feels supported by her partner will manage the demands that this new situation entails better and will adapt with greater confidence to the role of mother.

Self-efficacy in breastfeeding has been related to planned pregnancies [35] and to partner support in the early postpartum period [36]. Thus, it seems logical that support maintained from such early stages as planning for pregnancy is related to self-efficacy in breastfeeding beyond the first 6 weeks postpartum. The security and confidence that comes from feeling supported would increase perceived self-efficacy [37]which could explain the triangle of significant connections between partner support, success in breastfeeding and parental self-efficacy.

In terms of lifestyle, smoking is associated with a significantly lower quality of life in its physical dimension when the scores are adjusted for the variables under study. Smoking during pregnancy, in addition to negatively affecting the physical health of the woman after childbirth, is associated with postpartum depression [38] which would affect the physical and mental component of quality of life. All women are aware of the negative effect of smoking during pregnancy and postpartum on the fetus and infant, and therefore not being able to stop smoking for the benefit of the health of their child or taking up smoking again after childbirth is related to maternal anxiety, stress and lower self-confidence [39].

The holistic concept and deep interaction between physical and mental health are well known, which would explain why problems that affect physical health after childbirth such as dyspareunia, urinary incontinence, perineal pain or sleep deprivation [7] affect the mental component of quality of life, increasing the presence of depressive thoughts or symptoms. In turn, the presence of depressive thoughts or symptoms reduce the perception of physical well-being. This would explain the relationship between the perception of physical and mental well-being with an activation-reactivation cycle.

It is the responsibility of health services to provide quality care, focused on women, which is timely, appropriate, respectful and guided by the best scientific evidence available [3]. Furthermore, the midwife, due to her qualifications, proximity and involvement in the effective and satisfactory recovery of women after childbirth [40–42]is the professional who is most interested in understanding how the different factors involved in quality of life affect and interrelate with each other [9]in order to design interventions aimed at achieving optimal recovery after childbirth.

Thus, these results open up two avenues for reflection. One, the need to identify women with depressive symptoms in order to care for them as early as possible and intervene in those variables related to depression. To identify the women who are susceptible to depression, it is necessary to have reliable, easy-to-use and accessible tools that allow early detection [16, 43]and a rapid, standardized action strategy to be able to resolve or reverse the situation in an agile and satisfactory manner.

The other avenue for reflection, related to the first, is how to encourage support from the partner. In the health services, perhaps the only way is to involve the partner more in all the processes related to the sexual and reproductive health of the woman, from the choice of a contraceptive method, as a responsibility of both members of the couple, the preconception consultation, pregnancy, postpartum, and so on. In the long term, it is possible that the concept that sexual and reproductive health is a shared matter will be assumed; of course as long as the woman, the object of our care, agrees with and participates in this decision. Normalizing the involvement, participation and responsibility of the partner in the care of the children is a path that is being worked on with policies of conciliation and paternity leave, but there is still

a long way to go and our healthcare field does not cover it all.

Healthcare education aimed at women, their environment and society in general aimed at reducing social pressure on aesthetic standards, what is beautiful, and normalizing the physical changes that pregnancy and childbirth involve can, in the long term, reduce the expectations of the woman and her partner in terms of their body satisfaction. We should insist that physical recovery after childbirth is a long process, which requires time for the body to recover. We should work on a more realistic management of social expectations regarding motherhood, nowadays associated with a gratifying and highly romanticized state that leads to disappointment, to the harsh reality that is not easy to accept. Empowering women in their capacity, their beauty and their pleasure does not seem an easy task in this society, in which the value of image has so much weight; but our attitude and professional competence can be a step in the right direction.

Regarding the limitations of the study, we might consider that the sample was too small to assess such a wide range of variables, but we think that the diversity of participants was wide and representative of the society studied [44]except in the representation of the immigrant population, which is poorly represented in the sample studied due to the language barrier. However, measures have been taken to avoid introducing bias: the overwhelming majority of the women were selected by midwives belonging to public health centres located in various population areas, both rural and urban, and of different socioeconomic and social characteristics. Given that the women belonging to the snowball sample were referred by these same women, we can assume that they will come from equally varied backgrounds. Like any study that involves the involvement of participants to answer a questionnaire, it suffers from the bias that women who are more proactive, more interested or more sensitive respond more. In this sense, the online questionnaire offers the necessary privacy that could favor the participation of women who are more reserved or reluctant to share their situation; as well as the participation of busier women or those who are less willing to respond due to lack of time, since the online questionnaire can be answered at any time and in any place. On the other hand, there are so many variables involved in the quality of life that it is possible to have omitted the study of some of them; in order to try to avoid this, a previous study on the most relevant variables for health and well-being in the postpartum period in the Spanish population was used as a reference [16].

Strengths of the study include the high number of variables measured simultaneously and the use of subject-specific tools to measure variables and the hierarchical use of analytical statistics to investigate the relationship between variables [33]. The use of structural equation models has allowed the results obtained so far through partial models to be integrated into a single estimate, resulting in a more complete view of the aspects and pathways that determine quality of life during the postpartum period. In addition to the aforementioned ease of response provided by the online questionnaire, it is also worth adding the reduction of social-desirability bias.

We can conclude that this study provides a graphical representation that depicts the relationships between the different variables involved in quality of life after childbirth, which contributes to facilitating a global understanding of the entire process [45]. This knowledge forces us, as midwives, to review perinatal and postpartum care with the aim of paying special attention to depressive symptoms and partner support [33].

Acknowledgements
We would like to thank all the women who participated in the study for giving us their time and some of their privacy at such a vulnerable time in their lives as the postpartum period. We would like to thank all the midwives at primary care centers who are involved and committed to improving the care and health of women. And of course, we would like to thank the institutions that trust in our project and support us, Osakidetza and Biobizkaia. The ema.Q group is made up of Sonia Alvarez, Mónica Blas, Inés Cabeza, Itziar Estalella, Kata Legarra, Gorane Lozano, Amaia Maquibar, David Moreno-López, Mª Jesús Mulas, Mercedes Sáenz de Santamaría, Jesús Sánchez.

Author contributions
Conception of the idea for the study: IA and CP. Development of the protocol, organization and funding: IA, CP, ME, PB. Clinical committee responsible for the ítems selection: IA, CP, Ema.Q Group. Primary care clinical committee responsible for the recruitment work: IA, CP, Ema.Q Group. Coordinating centre: ME. Statistical analysis, writing of the manuscript: IA, ME, PB, AA. All the authors have read the draft critically, to make contributions, and have approved the final text.

Funding
This study has been funded by Instituto de Salud Carlos III (ISCIII) through the project "PI20/00899" and co-funded by the European Union. It also received funding from the Biobizkaia Health Research Institute (file number BB/I/LIB/24/010).

Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
The Basque Clinical Ethics Committee (file number PI2019110) approved the study in accordance with the relevant guidelines and regulations. Prior to participating in the study, all the participating women were informed of the study and an informed consent was provided by all the participants for this study. The consent and information sheet had been previously approved by the Ethics Committee. This study was conducted in compliance with the Declaration of Helsinki, which governs research involving human subjects and/or the use of human data.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 10 February 2025 / Accepted: 4 June 2025
Published online: 02 July 2025

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